A logistic regression model was constructed to predict the probability of detecting the Philippine Mouse-Deer (*Tragulus nigricans*) in Balabac, Palawan. The model utilized Average (Observer) Distance, Sampling Effort, and Habitat Code as independent variables.

```
## Call:
## glm(formula = Detected ~ AverageDistance + Effort + HabitatCode,
##     family = binomial(link = "logit"), data = clean_data)
##
## Coefficients:
##                   Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -19.57441 3525.85824  -0.006    0.996
## AverageDistance    0.16806    0.12985   1.294    0.196
## Effort             0.05477    1.10880   0.049    0.961
## HabitatCode2      19.79372 3525.85719   0.006    0.996
## HabitatCode3      16.51916 3525.85736   0.005    0.996
## HabitatCode4      36.08148 7414.61276   0.005    0.996
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 26.287  on 18  degrees of freedom
## Residual deviance: 14.638  on 13  degrees of freedom
## AIC: 26.638
##
## Number of Fisher Scoring iterations: 17
```

```
exp(coef(mousedeer_data))
```

```
##      (Intercept) AverageDistance          Effort    HabitatCode2    HabitatCode3
##     3.154568e-09    1.183003e+00    1.056302e+00    3.947325e+08    1.493411e+07
##     HabitatCode4
##     4.677210e+15
```

Figure 1. Summary Table of Logistic Regression Analysis

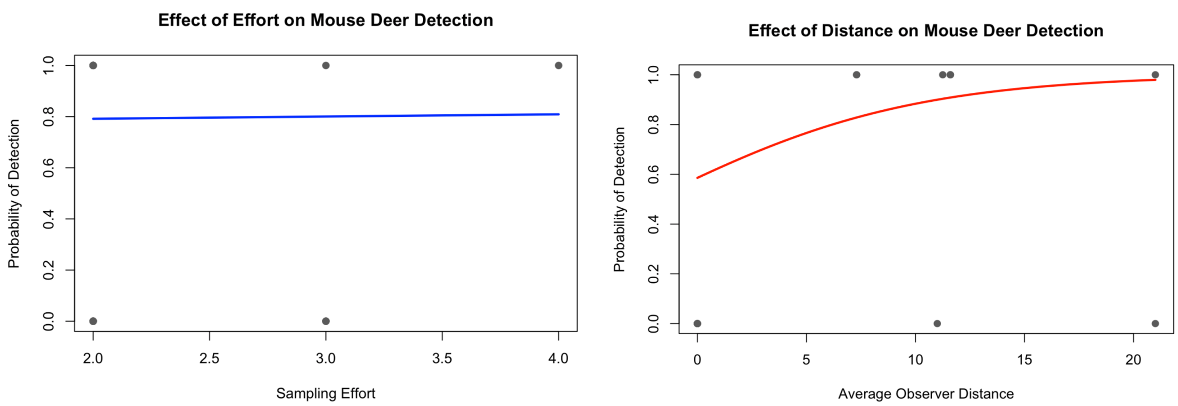

Figure 2a and 2b. Multiple Logistic Regression Plot with Habitat Code 1 as baseline

Interpretation:

Mouse-deer detection is found to be significantly higher in Habitat Codes 2, 3, and 4 compared to the Habitat Code 1 with the positive and high habitat estimate calculations (Figure 1).

Figure 2a tells us that as there is an increase in sampling effort, there is also an increased probability of detecting PH Mouse-deer by 5.6% per unit (Figure 1). However, the predicted probability curve is almost flat around 80% across the sampling effort x-axis, this implies that varying the sampling effort does not practically change the probability of observing mouse-deer and remains consistently high regardless.

For the average distance independent variable, the model predicts a 60% baseline chance of spotting mouse-deer at zero meters of observer distance within the area. But as the observer gets further away with constant sampling effort and habitat type, the chances increase by 18.3% (Figure 1 and 2b). This suggests that greater distances may allow broader and less obstructed field of view to detect mouse-deers without spooking them away.